Chapter 4, Problem 82P

Problem

An automobile battery has an open circuit voltage of 14.7 V which drops to 12 V when connected to two 65-W headlights. What is the resistance of the headlights and the value of the internal resistance of the battery?

Step-by-step solution

Step 1 of 4

The value of the Thevenin's voltage is equal to the open circuit voltage. That is,

$V_{\mathrm{Th}} = 14.7\ \mathrm{V}$.

Calculate the value of the current passing through $65\ \mathrm{W}$ load.

$$\begin{aligned} I &= \frac{65}{12} \\ &= 5.4167\ \mathrm{A} \end{aligned}$$

Therefore, the value of the current I is $5.4167\ \mathrm{A}$.

Step 2 of 4

Calculate the value of the load resistance R_L.

$$R_L = \frac{V}{I}$$

Substitute $5.4167\ \mathrm{A}$ for I and $12\ \mathrm{V}$ for V.

$$\begin{aligned} R_L &= \frac{12}{5.4167} \\ &= 2.215\ \Omega \end{aligned}$$

Therefore, the value of the load resistance R_L is $\boxed{2.215\ \Omega}$.

Step 3 of 4

Draw the Thevenin's equivalent circuit diagram including the load.

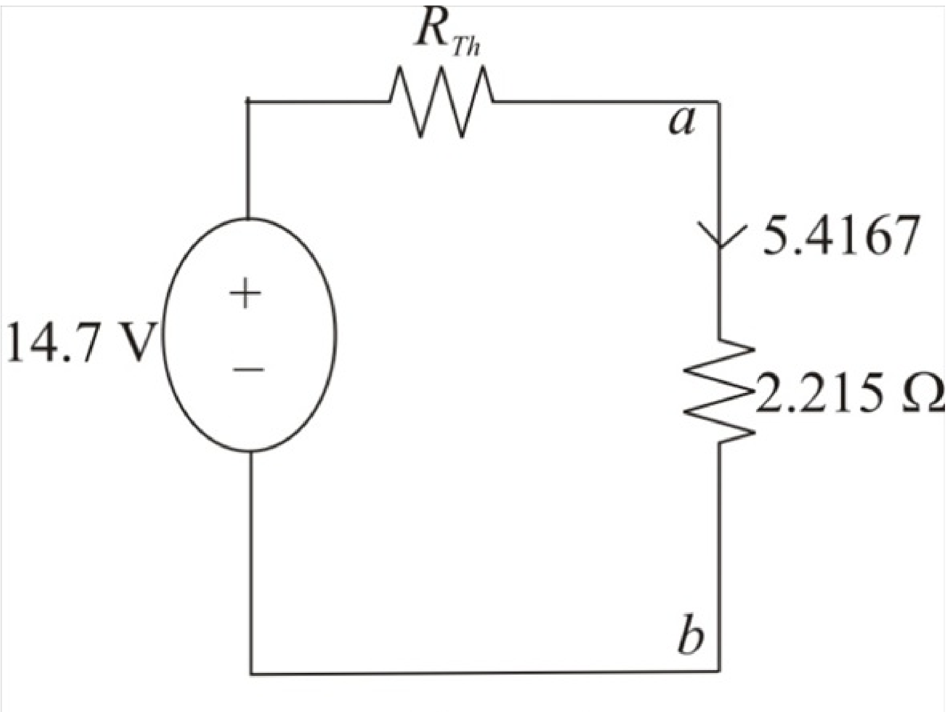

Figure 1

Step 4 of 4

Apply Kirchhoff's voltage law in the circuit.

$$-14.7+(5.4167)(R_{\text{Th}}+2.215)=0$$
$$(5.4167)(R_{\text{Th}}+2.215)=14.7$$
$$R_{\text{Th}}+2.215=\frac{14.7}{5.4167}$$
$$R_{\text{Th}}+2.215=2.7138$$
$$\begin{aligned}R_{\text{Th}}&=2.7138-2.215\\&=0.4988\\&\approx 0.5\ \Omega\end{aligned}$$

This value of the Thevenin's resistance is the internal resistance of the circuit.

Therefore, the value of the internal resistance is $\boxed{0.5\ \Omega}$.